Digital Signal Processing Course #CSE431
Lecture # 02 Topic: Classification of Signals

- Continuous-time signal vs. discrete-time signal
- Analog vs. Digital signal
- Periodic vs. aperiodic
- Causal vs. non-causal
- Even vs. odd
- Energy signal vs. power signal
- Deterministic vs. random
- Right-handed vs. left-handed
- Finite vs. infinite length (duration) signal

(i) Continuous-Time vs. Discrete-Time

A signal is said to be continuous when it is defined for all instants of time.
A signal is said to be discrete when it is defined at only discrete instants of time.
As the names suggest, this classification is determined by whether or not the time axis (x-axis) is *discrete* (countable) or *continuous* (Figure 1).

$$\text{CT signal: } x(t) = a\cos(wt + \theta)$$

$$\text{DT signal: } x[n] = x(nT) = \{x(0), x(T), \ldots, x(N-1)T\}, \quad n = 0, \pm 1, \pm 2, \ldots$$

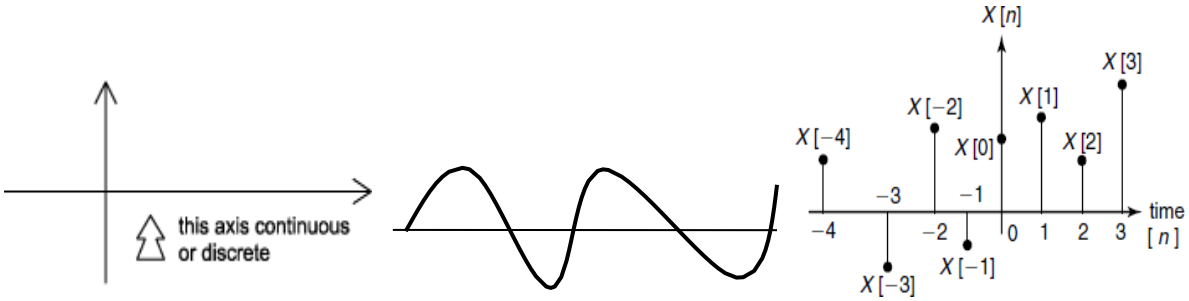


Figure 1

Examples:

- Continuous time signals (**t->time, x->signal, x(t) -> CT signal**)
  - Most of the signals in the physical world are CT signals—E.g. voltage & current, pressure, temperature, velocity, etc.
- Discrete-time signals in nature: (**n->time, x->signal, x[n] -> DT signal)**
  - DNA base sequence
  - Population of the nth generation of certain species

(ii) Analog vs. Digital

Analog corresponds to a continuous amplitude (y-axis) and time (x-axis), while digital corresponds to a discrete amplitude (y-axis) and time (x-axis). (Figure 2).

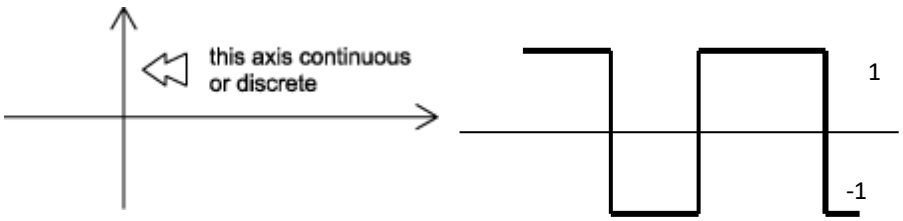


Figure 2

A Digital signal means a discrete signal with quantized (finite) amplitude values. For example, there are 2^8=256 gray scale levels in an 8-bit digital image.

Examples:

- Analog: signals in nature, such as a speech signal
- Digital: binary sequence, the transmission of data in a computer

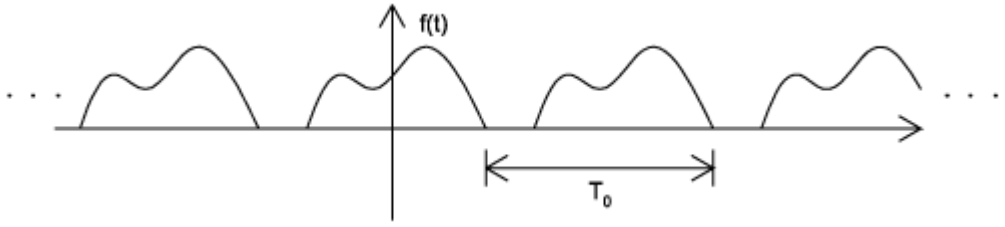
(iii) Periodic vs. Aperiodic (non-periodic)

A signal is said to be periodic if it satisfies the condition x(t) = x(t + T), where T = fundamental time period, 1/T = f = fundamental frequency.

A discrete-time signal x[n] is said to be periodic if it satisfies the following condition

$$x[n] = x[n+T], \text{ for all integer n}.$$

(a) A periodic signal with period T_0



(b) An aperiodic signal

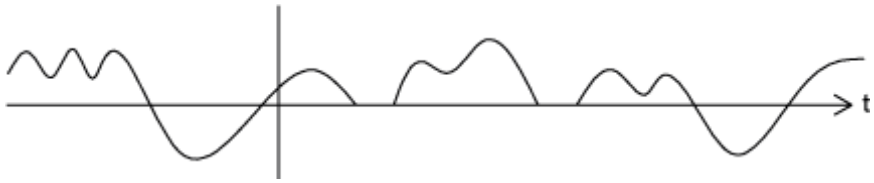


Figure 3

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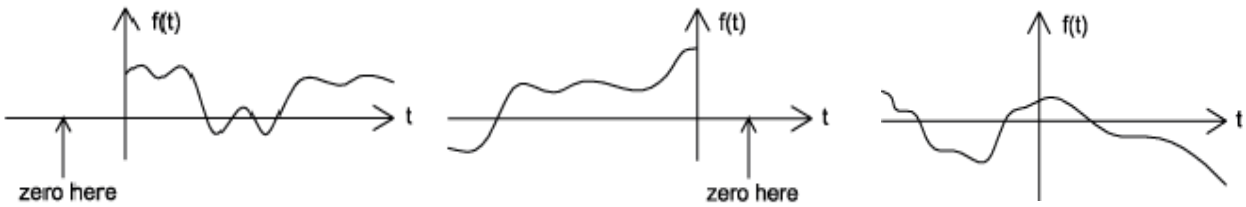
(iv) Causal vs. Anticausal or non-causal

Causal signals are signals that are zero for all negative time, while *anticausals* are signals that are zero for all positive time or nonzero values in both positive and negative time (Figure 4). A signal $x[n]$ is *causal* if $x[n] = 0$ for all $n < 0$. It is *anti-causal* if $x[n] = 0$ for all $n > 0$.

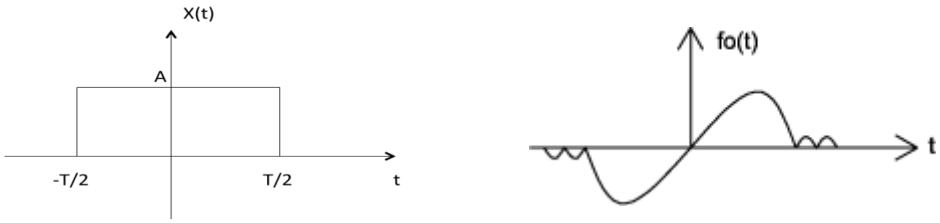


(a) A causal signal **(b) An anticausal or non-causal signal**

Figure 4

(v) Even vs. Odd

A signal is said to be even when it satisfies the condition x (t) = x (− t). Even signals can be easily spotted as they are *symmetric* around the vertical axis. An *odd signal*, on the other hand, is a signal *x* such that x(t) = − x(−t).

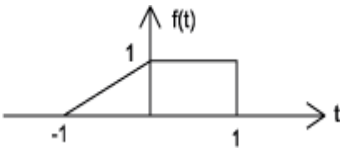


(a) An even signal (b) An odd signal

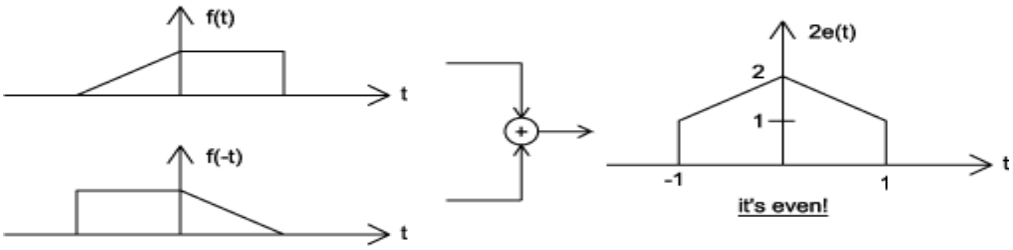
A function whose even part is zero is odd and a function whose odd part is zero is even.

Even function: $x_e(t) = \dfrac{x(t) + x(-t)}{2}$, Odd function: $x_o(t) = \dfrac{x(t) - x(-t)}{2}$

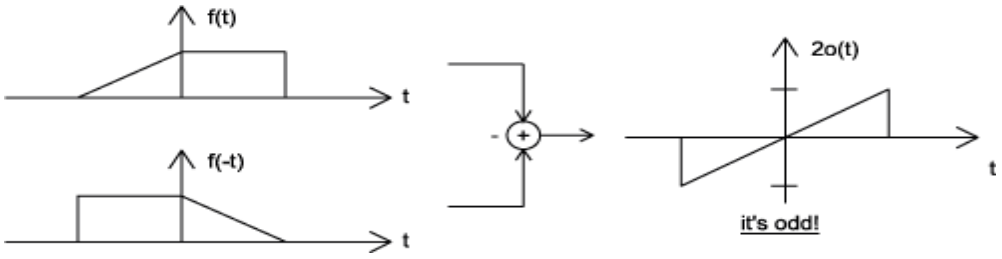
Example 1(a) The signal we will decompose using odd-even decomposition



(b) Even part: *e*(*t*) = ½ (*f*(*t*) +*f*(−*t*))



(c) Odd part: *o*(*t*) = ½ (*f*(*t*) −*f*(−*t*))



(d) Check: *e*(*t*) +*o*(*t*) =*f*(*t*)

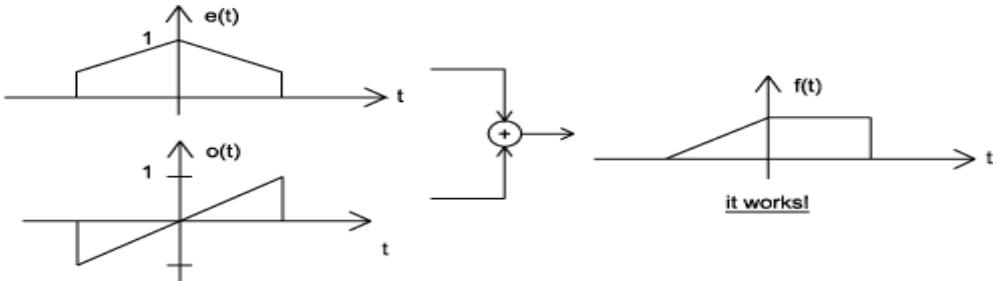


Figure 6

(vi) Deterministic vs. Random

A *deterministic signal* is a signal in which each value of the signal is fixed and can be determined by a mathematical expression, rule, or table. Because of this, the future values of the signal can be calculated from past values with complete confidence. On the other hand, a *random signal* has a lot of uncertainty about its behavior. The future values of a random signal cannot be accurately predicted and can usually only be guessed based on the averages of sets of signals (Figure 7).

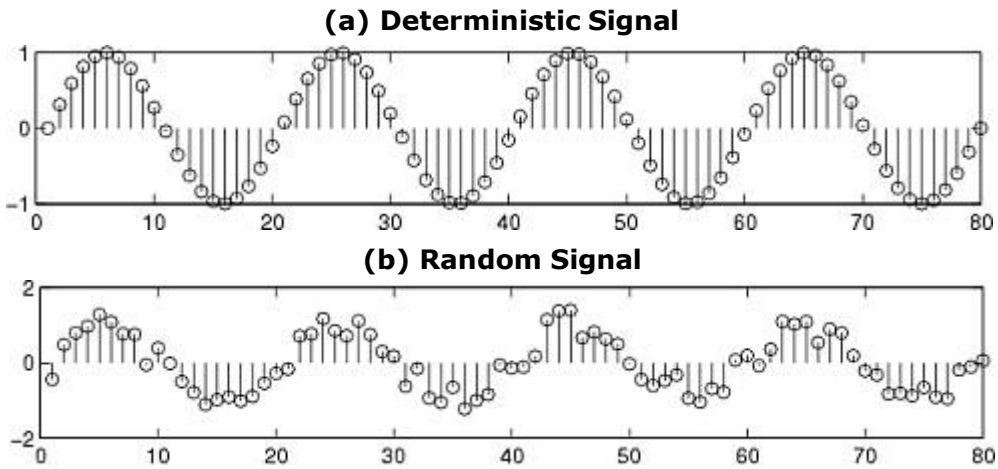
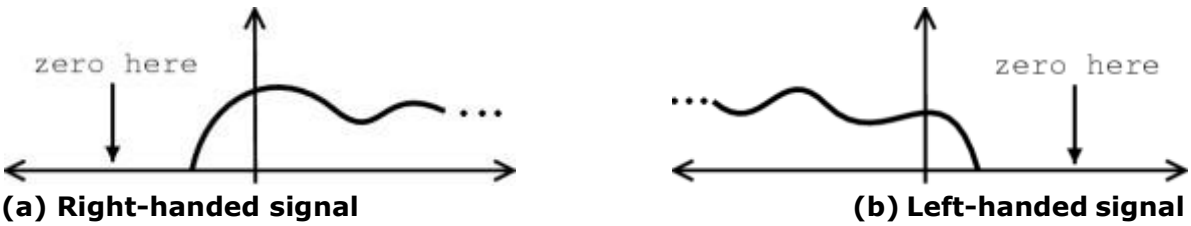


Figure 7

(vii) Right-handed vs. Left-Handed

A *right-handed* signal and a *left-handed* signal are those signals whose value is zero between a given variable and positive or negative infinity. Mathematically speaking, a right-handed signal is defined as any signal where $f(t) = 0$ for t<t_1<∞, and a left-handed signal is defined as any signal where $f(t) = 0$ for $t > t_1 > -\infty$. See (Figure 8) for an example. Both figures "begin" at t_1 and then extend to positive or negative infinity with mainly nonzero values.



(a) Right-handed signal **(b) Left-handed signal**

Figure 8

(viii) Finite vs. Infinite Length (duration)

As the name applies, signals can be characterized as to whether they have a finite or infinite length set of values. Most finite-length signals are used when dealing with discrete-time signals or a given sequence of values. Mathematically speaking, $f(t)$ is a *finite-length signal* if it is *nonzero* over a finite interval $t_1 < f(t) < t_2$ where $t_1 > -\infty$ and $t_2 < \infty$. An example can be seen in Figure 9. Similarly, an *infinite-length signal*, $f(t)$, is defined as nonzero overall real numbers: $\infty \leq f(t) \leq -\infty$.

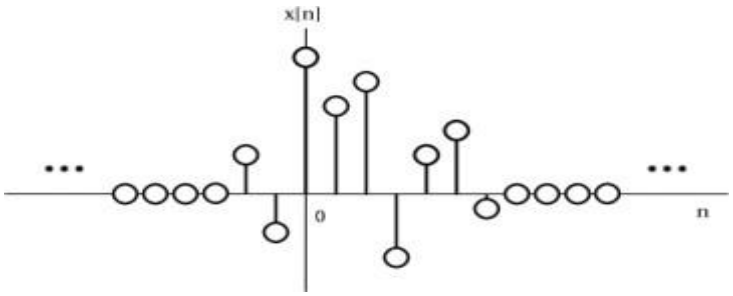


Figure 9: Finite-Length Signal. Note that it only has nonzero values on a set, finite interval.

First, we divide all signals into two classes: those that are of finite duration and those that are of infinite duration. A signal $x[n]$ is of fi*nite duration* if there exists two integers $\infty < N1 <= N2 < \infty$, such that $x[n] = 0$ only for $N1 <= n <= N2$. Otherwise, it is of *infinite duration*.

General rule:

- A signal cannot be both an energy and a power signal.
- A signal may be neither energy nor a power signal.
- All periodic signals are power signals (but not all non–periodic signals are energy signals).
- Any signal f that has limited amplitude ($|f| < \infty$) and is time limited ($f = 0$ for $|t| > t0$) is an energy signal.
- The square root of the average power of a power signal is called the RMS value.
- Periodic and random signals are power signals. Signals that are both deterministic and non-periodic are energy signals.